CS173: Discrete Structures
Problem Set 2

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1. Prove each of the following statements for all propositions φ, ψ, ξ, and χ.

   (a) $\vdash (\neg\varphi \to \varphi) \to \varphi$ — Consequentia Mirabilis

   Proof. *Let φ be an arbitrary proposition.*

   To prove $(\neg\varphi \to \varphi) \to \varphi$ it suffices to show $(\neg\varphi \to \varphi) \vdash \varphi$, by the deduction rule.

   Assume $(\neg\varphi \to \varphi)$.
   To prove φ, we use reductio ad absurdum.

   Assume $\neg\varphi$.
   By the assumption $(\neg\varphi \to \varphi)$, we have φ by modus ponens.
   However, we also have $\neg\varphi$ from our assumption.
   Thus we have both φ and $\neg\varphi$, which is a contradiction.

   Therefore, by reductio ad absurdum, φ holds.

   Therefore, by the deduction rule, $(\neg\varphi \to \varphi) \to \varphi$. $\square$

   (b) $\varphi \vdash \varphi \vee \psi$ — Disjunction Introduction

   Proof. *Let φ and ψ be arbitrary propositions. Assume φ.*

   To prove $\varphi \vee \psi$, we use reductio ad absurdum.

   Assume $\neg(\varphi \vee \psi)$.

   $$\neg(\varphi \vee \psi) \equiv \neg\varphi \wedge \neg\psi \qquad \text{(By De Morgan's law)}$$

   We know have $\neg\varphi \wedge \neg\psi$.
   By conjunction elimination, we have $\neg\varphi$.
   However, we originally assumed φ.
   Thus we have both φ and $\neg\varphi$, which is a contradiction.

   Therefore, by reductio ad absurdum, $\varphi \vee \psi$ holds. $\square$

(c) $(\varphi \vee \psi), (\varphi \to \xi), (\psi \to \xi) \vdash \xi$ Disjunction Elimination

Proof. *Let* $\varphi, \psi,$ *and* ξ *be arbitrary propositions. Assume* $(\varphi \vee \psi)$*,* $(\varphi \to \xi)$*, and* $(\psi \to \xi)$*.*

To prove ξ*, we use reductio ad absurdum.*

> *Assume* $\neg\xi$*.*
> *By the assumption* $(\varphi \to \xi)$ *and* $\neg\xi$*, we have* $\neg\varphi$ *by modus tollens.*
> *By the assumption* $(\psi \to \xi)$ *and* $\neg\xi$*, we have* $\neg\psi$ *by modus tollens.*
> *By conjunction introduction, we have* $\neg\varphi \wedge \neg\psi$*:*
>
> $$\neg\varphi \wedge \neg\psi \equiv \neg(\varphi \vee \psi) \qquad \text{(By De Morgan's law)}$$
>
> *Thus we have* $\neg(\varphi \vee \psi)$*.*
> *However, we originally assumed* $(\varphi \vee \psi)$*.*
> *Thus we have both* $(\varphi \vee \psi)$ *and* $\neg(\varphi \vee \psi)$*, which is a contradiction.*

Therefore, by reductio ad absurdum, ξ *holds.* □

(d) $\varphi, \neg\varphi \vdash \psi$ Ex Falso Quodlibet

Proof. *Let* φ *and* ψ *be arbitrary propositions. Assume* φ *and* $\neg\varphi$*.*

By conjunction introduction, we have $\varphi \wedge \neg\varphi$*.*

$$\begin{aligned} \varphi \wedge \neg\varphi &\equiv \neg\varphi \wedge \varphi && \text{(By Commutativity)} \\ &\equiv \bot && \text{(By Complement)} \end{aligned}$$

Thus we have $\bot$*.*

To prove ψ*, we use reductio ad absurdum.*

> *Assume* $\neg\psi$*.*
> *We have previously proven* $\top$*. Furthermore, we have also proven* $\top \equiv \neg\bot$
> *We have both* $\bot$ *and* $\neg\bot$*, which is a contradiction.*

Therefore, by reductio ad absurdum, ψ *holds.* □

(e) $(\varphi \vee \psi), \neg\varphi \vdash \psi$ Disjunctive Syllogism

Proof. *Let φ and ψ be arbitrary propositions. Assume $\varphi \vee \psi$ and also assume $\neg\varphi$. By conjunction introduction, we have $(\varphi \vee \psi) \wedge \neg\varphi$:*

$$
\begin{aligned}
(\varphi \vee \psi) \wedge \neg\varphi &\equiv \neg\varphi \wedge (\varphi \vee \psi) && \text{(By Commutativity)} \\
&\equiv (\neg\varphi \wedge \varphi) \vee (\neg\varphi \wedge \psi) && \text{(By Distributivity)} \\
&\equiv \bot \vee (\neg\varphi \wedge \psi) && \text{(By Complement)} \\
&\equiv \neg\varphi \wedge \psi && \text{(By Identity)} \\
&\equiv \psi \wedge \neg\varphi && \text{(By Commutativity)}
\end{aligned}
$$

By conjunction elimination, we have ψ.
Therefore, ψ holds. □

(f) $(\varphi \to \xi), (\psi \to \chi), (\varphi \vee \psi) \vdash (\xi \vee \chi)$ Constructive Dilemma

Proof. *Let $\varphi, \psi, \xi,$ and χ be arbitrary propositions. Assume $(\varphi \to \xi)$, $(\psi \to \chi)$, and $(\varphi \vee \psi)$.*

To prove $\xi \vee \chi$, we use reductio ad absurdum.

Assume $\neg(\xi \vee \chi)$.

$$\neg(\xi \vee \chi) \equiv \neg\xi \wedge \neg\chi \qquad \text{(By De Morgan's law)}$$

By conjunction elimination, we have $\neg\xi$ and $\neg\chi$.
By the assumption $(\varphi \to \xi)$ and $\neg\xi$, we have $\neg\varphi$ by modus tollens.
By the assumption $(\psi \to \chi)$ and $\neg\chi$, we have $\neg\psi$ by modus tollens.
By conjunction introduction, we have $\neg\varphi \wedge \neg\psi$.

$$\neg\varphi \wedge \neg\psi \equiv \neg(\varphi \vee \psi) \qquad \text{(By De Morgan's law)}$$

Thus we have $\neg(\varphi \vee \psi)$.
However, we originally assumed $(\varphi \vee \psi)$.
Thus we have both $(\varphi \vee \psi)$ and $\neg(\varphi \vee \psi)$, which is a contradiction.

Therefore, by reductio ad absurdum, $\xi \vee \chi$ holds. □

2. Prove each of the following statements.

(a) Consider a universe of discourse containing all human beings who have ever existed or been mentioned in literature. Let $\boldsymbol{\mu}$ and $\boldsymbol{\omega}$ be unary predicates.
Prove that $\forall \boldsymbol{x}(\boldsymbol{\mu}(\boldsymbol{x}) \to \boldsymbol{\omega}(\boldsymbol{x})), \boldsymbol{\mu}(\text{Socrates}) \vdash \boldsymbol{\omega}(\text{Socrates})$.

Proof. *Assume* $\forall \boldsymbol{x}(\boldsymbol{\mu}(\boldsymbol{x}) \to \boldsymbol{\omega}(\boldsymbol{x}))$ *and* $\boldsymbol{\mu}(\textit{Socrates})$.

By universal elimination on $\forall \boldsymbol{x}(\boldsymbol{\mu}(\boldsymbol{x}) \to \boldsymbol{\omega}(\boldsymbol{x}))$ *with* $\boldsymbol{x} = \textit{Socrates}$, *we have* $\boldsymbol{\mu}(\textit{Socrates}) \to \boldsymbol{\omega}(\textit{Socrates})$.

By the assumption $\boldsymbol{\mu}(\textit{Socrates})$ *and* $\boldsymbol{\mu}(\textit{Socrates}) \to \boldsymbol{\omega}(\textit{Socrates})$, *we have* $\boldsymbol{\omega}(\textit{Socrates})$ *by modus ponens.*

Therefore, $\boldsymbol{\omega}(\textit{Socrates})$ *holds.* □

(b) Consider a non-empty universe of discourse consisting of all persons inhabiting New Port City, Japan in the year 2029 ad. Let $\boldsymbol{\alpha}$ and $\boldsymbol{\gamma}$ be unary predicates.
Prove that $\forall \boldsymbol{x}(\neg\gamma(\boldsymbol{x}) \to \boldsymbol{\alpha}(\boldsymbol{x})), \neg\boldsymbol{\alpha}(\text{Kusanagi}) \vdash \exists \boldsymbol{x}(\gamma(\boldsymbol{x}))$.

Proof. *Assume* $\forall \boldsymbol{x}(\neg\gamma(\boldsymbol{x}) \to \boldsymbol{\alpha}(\boldsymbol{x}))$ *and* $\neg\boldsymbol{\alpha}(\textit{Kusanagi})$.

For clarity, let $\boldsymbol{K} = \textit{Kusanagi}$.

By universal elimination on $\forall \boldsymbol{x}(\neg\gamma(\boldsymbol{x}) \to \boldsymbol{\alpha}(\boldsymbol{x}))$ *with* $\boldsymbol{x} = \boldsymbol{K}$, *we have* $\neg\gamma(\boldsymbol{K}) \to \boldsymbol{\alpha}(\boldsymbol{K})$.

By the assumption $\neg\boldsymbol{\alpha}(\boldsymbol{K})$ *and* $\neg\gamma(\boldsymbol{K}) \to \boldsymbol{\alpha}(\boldsymbol{K})$, *we have* $\neg(\neg\gamma(\boldsymbol{K}))$ *by modus tollens.*

$$\neg(\neg\gamma(\boldsymbol{K})) \equiv \gamma(\boldsymbol{K}) \qquad \textit{(By Double Negation)}$$

Thus we have $\boldsymbol{\gamma}(\boldsymbol{K})$.

By existential introduction on $\gamma(\boldsymbol{K})$, *we have* $\exists \boldsymbol{x}(\gamma(\boldsymbol{x}))$.

Therefore, $\exists \boldsymbol{x}(\gamma(\boldsymbol{x}))$ *holds.* □

(c) Let $\boldsymbol{\mathcal{L}}$ be a binary predicate and show that $\vdash \neg\exists \boldsymbol{x}\forall \boldsymbol{y}(\boldsymbol{\mathcal{L}}(\boldsymbol{x},\boldsymbol{y}) \leftrightarrow \neg\boldsymbol{\mathcal{L}}(\boldsymbol{y},\boldsymbol{y}))$.

Proof. *To prove* $\neg\exists \boldsymbol{x}\forall \boldsymbol{y}(\boldsymbol{\mathcal{L}}(\boldsymbol{x},\boldsymbol{y}) \leftrightarrow \neg\boldsymbol{\mathcal{L}}(\boldsymbol{y},\boldsymbol{y}))$*, we use reductio ad absurdum.*

Assume $\neg(\neg\exists \boldsymbol{x}\forall \boldsymbol{y}(\boldsymbol{\mathcal{L}}(\boldsymbol{x},\boldsymbol{y}) \leftrightarrow \neg\boldsymbol{\mathcal{L}}(\boldsymbol{y},\boldsymbol{y})))$*, which, by double negation, equates to:*

$$\exists \boldsymbol{x}\forall \boldsymbol{y}(\boldsymbol{\mathcal{L}}(\boldsymbol{x},\boldsymbol{y}) \leftrightarrow \neg\boldsymbol{\mathcal{L}}(\boldsymbol{y},\boldsymbol{y}))$$

By existential elimination, for some new term $\boldsymbol{a}$*, we have*
$\forall \boldsymbol{y}(\boldsymbol{\mathcal{L}}(\boldsymbol{a},\boldsymbol{y}) \leftrightarrow \neg\boldsymbol{\mathcal{L}}(\boldsymbol{y},\boldsymbol{y}))$.
By universal elimination with $\boldsymbol{y} = \boldsymbol{a}$*, we have* $\boldsymbol{\mathcal{L}}(\boldsymbol{a},\boldsymbol{a}) \leftrightarrow \neg\boldsymbol{\mathcal{L}}(\boldsymbol{a},\boldsymbol{a})$.
Which, by biconditional disintegration, equates to:

$$(\boldsymbol{\mathcal{L}}(\boldsymbol{a},\boldsymbol{a}) \to \neg\boldsymbol{\mathcal{L}}(\boldsymbol{a},\boldsymbol{a})) \wedge (\neg\boldsymbol{\mathcal{L}}(\boldsymbol{a},\boldsymbol{a}) \to \boldsymbol{\mathcal{L}}(\boldsymbol{a},\boldsymbol{a}))$$

By conjunction elimination, we have $\boldsymbol{\mathcal{L}}(\boldsymbol{a},\boldsymbol{a}) \to \neg\boldsymbol{\mathcal{L}}(\boldsymbol{a},\boldsymbol{a})$ *and*
$\neg\boldsymbol{\mathcal{L}}(\boldsymbol{a},\boldsymbol{a}) \to \boldsymbol{\mathcal{L}}(\boldsymbol{a},\boldsymbol{a})$.
Consider $\boldsymbol{\mathcal{L}}(\boldsymbol{a},\boldsymbol{a}) \to \neg\boldsymbol{\mathcal{L}}(\boldsymbol{a},\boldsymbol{a})$*. By conditional disintegration:*

$$\begin{aligned}
\boldsymbol{\mathcal{L}}(\boldsymbol{a},\boldsymbol{a}) \to \neg\boldsymbol{\mathcal{L}}(\boldsymbol{a},\boldsymbol{a}) &\equiv \neg\boldsymbol{\mathcal{L}}(\boldsymbol{a},\boldsymbol{a}) \vee \neg\boldsymbol{\mathcal{L}}(\boldsymbol{a},\boldsymbol{a}) && \textit{(By Conditional Disintegration)}\\
&\equiv \neg\boldsymbol{\mathcal{L}}(\boldsymbol{a},\boldsymbol{a}) && \textit{(By Idempotence)}
\end{aligned}$$

Thus we have $\neg\boldsymbol{\mathcal{L}}(\boldsymbol{a},\boldsymbol{a})$.
Consider $\neg\boldsymbol{\mathcal{L}}(\boldsymbol{a},\boldsymbol{a}) \to \boldsymbol{\mathcal{L}}(\boldsymbol{a},\boldsymbol{a})$*. By conditional disintegration:*

$$\begin{aligned}
\neg\boldsymbol{\mathcal{L}}(\boldsymbol{a},\boldsymbol{a}) \to \boldsymbol{\mathcal{L}}(\boldsymbol{a},\boldsymbol{a}) &\equiv \neg(\neg\boldsymbol{\mathcal{L}}(\boldsymbol{a},\boldsymbol{a})) \vee \boldsymbol{\mathcal{L}}(\boldsymbol{a},\boldsymbol{a}) && \textit{(By Conditional Disintegration)}\\
&\equiv \boldsymbol{\mathcal{L}}(\boldsymbol{a},\boldsymbol{a}) \vee \boldsymbol{\mathcal{L}}(\boldsymbol{a},\boldsymbol{a}) && \textit{(By Double Negation)}\\
&\equiv \boldsymbol{\mathcal{L}}(\boldsymbol{a},\boldsymbol{a}) && \textit{(By Idempotence)}
\end{aligned}$$

Thus we have $\boldsymbol{\mathcal{L}}(\boldsymbol{a},\boldsymbol{a})$.
We have both $\boldsymbol{\mathcal{L}}(\boldsymbol{a},\boldsymbol{a})$ *and* $\neg\boldsymbol{\mathcal{L}}(\boldsymbol{a},\boldsymbol{a})$*, which is a contradiction.*

Therefore, by reductio ad absurdum, $\neg\exists \boldsymbol{x}\forall \boldsymbol{y}(\boldsymbol{\mathcal{L}}(\boldsymbol{x},\boldsymbol{y}) \leftrightarrow \neg\boldsymbol{\mathcal{L}}(\boldsymbol{y},\boldsymbol{y}))$ *holds.* □

3. For this problem, we will be treating a statement that is not a proposition as if it is a proposition as a cautionary tale. Remember that, for the purposes of proof-writing, we do not yet know any facts about the natural numbers.

   Consider a universe of discourse consisting of the natural numbers. Let $\boldsymbol{\varphi}$ be the statement $\boldsymbol{\varphi} \to \forall \boldsymbol{x}(\boldsymbol{\pi}(\boldsymbol{x}))$.

   Prove that 4 is a prime number.

   Proof. *Let* $\boldsymbol{\varphi}$ *be the statement* $\boldsymbol{\varphi} \to \forall \boldsymbol{x}(\boldsymbol{\pi}(\boldsymbol{x}))$.

   By existential elimination, to prove $\boldsymbol{\pi}(\mathbf{4})$ *we will show* $\forall \boldsymbol{x}(\boldsymbol{\pi}(\boldsymbol{x}))$.

   By the deduction rule, to prove $\forall \boldsymbol{x}(\boldsymbol{\pi}(\boldsymbol{x}))$ *from* $\boldsymbol{\varphi} \to \forall \boldsymbol{x}(\boldsymbol{\pi}(\boldsymbol{x}))$, *we first show* $\boldsymbol{\varphi}$.

   We use reductio ad absurdum.

   Assume $\neg\boldsymbol{\varphi}$.
   By disjunction introduction we have $\neg\boldsymbol{\varphi} \vee \forall \boldsymbol{x}(\boldsymbol{\pi}(\boldsymbol{x}))$:

   $$\neg\boldsymbol{\varphi} \vee \forall \boldsymbol{x}(\boldsymbol{\pi}(\boldsymbol{x})) \equiv \boldsymbol{\varphi} \to \forall \boldsymbol{x}(\boldsymbol{\pi}(\boldsymbol{x})) \qquad \textit{(By Conditional Disintegration)}$$

   However, $\boldsymbol{\varphi}$ *is defined to be the statement* $\boldsymbol{\varphi} \to \forall \boldsymbol{x}(\boldsymbol{\pi}(\boldsymbol{x}))$.
   Thus we have $\boldsymbol{\varphi}$.
   We have both $\boldsymbol{\varphi}$ *and* $\neg\boldsymbol{\varphi}$, *which is a contradiction.*

   Therefore, by reductio ad absurdum, $\boldsymbol{\varphi}$ *holds.*

   By universal elimination with $\boldsymbol{x} = \mathbf{4}$, *we have* $\boldsymbol{\pi}(\mathbf{4})$.

   Therefore, 4 is a prime number. □